

MCCAMPBELL-PORTIER, TX, USA

WMO: 722369

Lat: 27.917N

Lon: 97.200W

Elev: 16

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	34.4	37.5	16.3	12.5	47.3	22.0	16.5	50.9	26.9	60.1	24.9	61.4	8.9	0	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.7	95.3	77.6	93.9	78.1	92.9	78.3	81.7	89.3	81.1	88.7	80.5	88.0	12.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.8	154.8	84.5	79.2	151.8	84.4	78.7	149.3	84.4	45.1	89.0	44.5	89.0	43.9	88.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.6	22.1	20.0	DB	28.3	99.9	4.8	2.4	24.8	101.6	22.0	103.0	19.3	104.3	15.9	106.0
			WB	25.3	82.3	4.3	1.1	22.3	83.0	19.8	83.7	17.4	84.3	14.2	85.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.3	57.0	61.7	66.8	73.3	78.7	83.7	85.8	86.2	82.9	76.4	66.2	59.9	(6)
(7)		DBStd	12.24	8.99	9.18	8.32	5.67	5.42	5.01	2.28	2.38	3.93	6.96	9.18	8.97	(7)
(8)		HDD50	89	38	17	9	0	0	0	0	0	0	0	7	18	(8)
(9)		HDD65	839	275	153	80	9	1	3	0	0	0	9	97	212	(9)
(10)		CDD50	8583	255	344	528	698	891	1012	1109	1123	988	819	492	324	(10)
(11)		CDD65	3860	27	60	135	257	427	565	644	658	538	364	132	53	(11)
(12)		CDH74	41718	92	168	504	1641	4217	7069	8385	8617	6277	3684	881	183	(12)
(13)		CDH80	17878	5	12	44	317	1325	3265	4161	4449	2823	1310	158	9	(13)
(14)	Wind	WSAvg	10.3	9.7	10.6	11.4	12.2	12.2	11.0	10.8	9.5	8.3	9.1	9.0	9.5	(14)
(15)	Precipitation	PrecAvg	34.3	2.1	2.1	2.6	2.1	3.3	2.6	3.1	2.0	5.9	4.2	2.8	1.7	(15)
(16)		PrecMax	50.0	6.3	5.3	5.4	8.6	12.4	8.9	13.8	7.2	16.6	14.1	10.0	6.0	(16)
(17)		PrecMin	18.2	0.0	0.1	0.1	0.2	0.0	0.0	0.0	0.0	1.3	0.0	0.0	0.3	(17)
(18)		PrecStd	9.5	1.8	1.7	1.7	2.0	3.3	2.2	3.9	1.8	3.7	3.9	2.7	1.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.9	81.5	84.2	90.7	92.4	97.1	96.1	98.8	97.2	92.4	86.9	81.3	(19)
(20)		MCWB	66.4	67.7	69.0	64.6	73.0	76.9	78.2	75.6	76.8	77.2	75.6	69.3	(20)	
(21)		2%	DB	76.8	78.7	80.9	85.7	89.9	93.3	94.1	95.4	93.8	90.3	83.9	78.8	(21)
(22)		MCWB	67.0	69.6	70.8	71.0	75.9	78.3	78.7	78.0	77.5	78.0	74.2	70.3	(22)	
(23)		5%	DB	73.4	76.1	79.1	82.7	88.1	91.7	93.0	94.5	92.2	88.5	81.6	75.7	(23)
(24)		MCWB	65.0	68.5	70.1	72.0	76.2	78.4	78.9	78.3	77.8	76.6	71.6	68.7	(24)	
(25)		10%	DB	71.6	73.1	77.0	81.4	86.4	90.7	91.8	93.1	90.8	86.5	79.2	73.0	(25)
(26)		MCWB	65.1	67.1	68.9	71.7	76.0	78.3	79.0	78.5	77.7	75.0	71.0	68.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.2	73.5	74.4	77.8	80.4	82.5	82.3	82.2	82.3	81.6	78.3	74.4	(27)
(28)		MCDWB	76.4	78.9	80.0	83.3	86.3	89.7	89.8	90.8	90.4	87.3	84.0	78.7	(28)	
(29)		2%	WB	70.5	72.0	73.3	76.2	79.2	81.3	81.4	81.2	81.2	80.2	76.2	72.9	(29)
(30)		MCDWB	73.8	76.1	78.3	81.4	84.8	88.6	89.4	89.1	88.8	86.3	80.5	76.4	(30)	
(31)		5%	WB	69.0	70.7	72.3	75.0	78.4	80.4	80.9	80.5	80.4	79.3	74.5	71.4	(31)
(32)		MCDWB	71.7	73.9	76.7	80.0	84.2	87.7	88.8	88.3	87.8	85.5	78.8	73.8	(32)	
(33)		10%	WB	66.5	69.4	71.1	74.1	77.7	79.8	80.3	80.1	79.8	78.2	73.1	69.7	(33)
(34)		MCDWB	69.5	71.8	75.1	78.8	83.6	87.2	88.1	87.9	87.0	84.3	77.7	72.5	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.7	16.8	15.5	15.0	13.5	13.1	12.8	14.7	15.3	17.2	18.0	17.8	(35)	
(36)		5% DB	MCDBR	20.7	16.6	13.8	15.6	14.1	13.6	14.0	16.4	16.5	15.4	17.4	17.5	(36)
(37)		MCWBR	13.0	9.7	7.0	7.5	4.9	4.0	4.1	4.7	5.5	6.5	9.7	11.6	(37)	
(38)		5% WB	MCDBR	15.6	13.8	12.5	11.4	10.6	11.7	12.0	13.4	13.2	13.4	15.2	15.4	(38)
(39)		MCWBR	12.2	9.9	7.1	6.2	4.4	4.1	3.9	4.2	5.0	6.9	9.8	12.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.361	0.367	0.387	0.429	0.471	0.448	0.465	0.449	0.439	0.393	0.378	0.349	(40)	
(41)		taud	2.476	2.461	2.402	2.291	2.205	2.333	2.303	2.346	2.337	2.462	2.453	2.525	(41)	
(42)		Ebn at Noon	279	287	287	276	263	268	263	267	267	275	271	279	(42)	
(43)		Edh at Noon	30	33	37	43	46	41	42	40	39	33	31	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	947	1119	1437	1711	1923	2121	2087	1996	1651	1438	1078	853	(44)	
(45)		RadStd	104	185	151	162	135	110	176	110	142	148	113	113	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146

Nomenclature: See separate page